

Vara Prasad

Software Developer & AI Engineer

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🌐 **Portfolio:** [portfolio-link]

Career Objective & Professional Summary

I am a passionate and dedicated software developer with expertise in full-stack web development, artificial intelligence, and machine learning. My technical journey spans across modern web technologies including React.js, Vue.js, Node.js, FastAPI, and Python, combined with extensive experience in database management using PostgreSQL, MySQL, and MongoDB. I specialize in building scalable, user-centric applications that solve real-world problems while maintaining high code quality and performance standards.

Currently leading the development of a Telugu Large Language Model (LLM) as part of a prestigious government-backed AI initiative, I bring strong project management skills, technical leadership experience, and a deep understanding of AI/ML workflows including data preprocessing, model training, fine-tuning, and deployment. My expertise extends to cloud technologies (AWS), containerization (Docker), and modern development practices including version control with Git, GitLab, and collaborative development workflows.

I am seeking opportunities to contribute to innovative software development teams where I can leverage my technical skills in web development, artificial intelligence, and system architecture to build impactful solutions. I thrive in collaborative environments and am committed to continuous learning, writing maintainable code, and delivering high-quality software products that make a meaningful difference.

Education

Bachelor of Technology in Electronics and Communication Engineering

B V Raju Institute Of Technology, Narsapur

📅 **Duration:** August 2023 – April 2027

🎓 **CGPA:** 8.89/10.0

Relevant Coursework & Academic Focus:

- **Digital Signal Processing:** Understanding of signal analysis, filtering, and processing techniques that complement my AI/ML work
- **Communication Systems:** Knowledge of data transmission, networking protocols, and system design principles
- **Embedded Systems:** Hardware-software integration, microcontroller programming, and IoT applications
- **Mathematics & Statistics:** Strong foundation in calculus, linear algebra, probability, and statistics essential for machine learning
- **Programming Fundamentals:** Algorithmic thinking, data structures, and computational problem-solving
- **Project Management:** Academic projects involving team collaboration, timeline management, and technical documentation

Academic Achievements:

- Consistent academic performance with CGPA of 8.89, demonstrating strong analytical and problem-solving abilities
 - Active participation in technical workshops, coding competitions, and collaborative projects
 - Integration of theoretical knowledge with practical programming skills through various academic and personal projects
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Professional Experience

Software Developer Intern

SOAI Hybrid | June 2025 – Present

 **Location:** Hyderabad, India

 **Government Partnership:** Telangana State AI Initiative

Project: Telugu Large Language Model (LLM) Development

Leadership & Team Management:

- **Team Leadership:** Successfully managing and coordinating a diverse cross-functional team of 200+ developers, data scientists, linguists, and technical contributors across multiple time zones and expertise areas
- **Project Coordination:** Implementing agile methodologies and project management best practices to ensure seamless collaboration, milestone tracking, and deliverable management
- **Stakeholder Communication:** Regular reporting and coordination with government officials, technical advisors, and project stakeholders to ensure alignment with policy objectives and technical requirements

- **Resource Management:** Allocating technical resources, managing development sprints, and ensuring optimal team productivity and morale

Technical Architecture & Implementation:

- **Data Pipeline Development:** Designing and implementing robust, scalable data pipelines for processing large-scale Telugu language datasets, including web scraping, text cleaning, normalization, and quality assurance workflows
- **Model Training Infrastructure:** Architecting distributed training systems using modern ML frameworks (TensorFlow, PyTorch) with GPU acceleration, managing computational resources, and optimizing training efficiency
- **Tokenization & Preprocessing:** Developing specialized tokenization strategies for Telugu language processing, handling character encoding, script variations, and linguistic nuances specific to Dravidian languages
- **Fine-tuning Workflows:** Implementing advanced fine-tuning techniques including transfer learning, domain adaptation, and parameter-efficient training methods to optimize model performance for regional language understanding

AI/ML Technical Expertise:

- **Neural Network Architecture:** Working with transformer architectures, attention mechanisms, and large-scale language modeling techniques
- **Dataset Curation:** Managing data collection, cleaning, and validation processes for Telugu language corpora, ensuring data quality and ethical sourcing
- **Model Evaluation:** Implementing comprehensive evaluation metrics, benchmarking against existing models, and continuous performance monitoring
- **Deployment Strategy:** Planning and executing model deployment pipelines for production-ready AI systems

Government & Policy Compliance:

- **Ethical AI Implementation:** Ensuring compliance with government AI ethics guidelines, data privacy regulations, and responsible AI development practices
- **Policy Alignment:** Working closely with Telangana government bodies to align technical development with state AI policy frameworks and strategic objectives
- **Documentation & Reporting:** Maintaining comprehensive technical documentation, progress reports, and compliance records for government oversight and transparency

Technical Impact & Innovation:

- Contributing to the advancement of regional language AI capabilities in India
- Developing innovative solutions for low-resource language processing challenges

- Building scalable AI infrastructure that can be adapted for other regional languages
 - Mentoring junior developers and contributing to knowledge sharing within the team
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Technical Projects

WealthMap - Financial Analytics Platform

Duration: May 2025 – June 2025

 **Technologies:** Vue.js, FastAPI, PostgreSQL, OpenLeaf Maps, Firebase, Google APIs, ATTOM API

Project Overview: WealthMap is a comprehensive financial analytics and investment research platform designed to provide real estate market insights, investment opportunities, and portfolio management tools. The platform serves two primary user roles: administrators who manage company profiles and employees who explore investment opportunities and market data.

Frontend Development (Vue.js):

- **Vue.js Architecture:** Built a responsive, component-based user interface using Vue 3 with Composition API, implementing reactive data binding, computed properties, and lifecycle hooks for optimal performance
- **State Management:** Implemented Vuex for centralized state management, handling user authentication states, application data, and API response caching
- **Responsive Design:** Created mobile-first responsive layouts using CSS Grid, Flexbox, and Vue's transition system for smooth user interactions across all device sizes
- **Component Library:** Developed reusable Vue components for data visualization, form handling, map integration, and user interface elements

Backend Development (FastAPI):

- **API Architecture:** Designed RESTful APIs using FastAPI with automatic OpenAPI documentation, implementing proper HTTP status codes, error handling, and request/response validation
- **Database Design:** Created efficient database schemas with proper normalization, indexing strategies, and relationship management for user data, company profiles, and investment records
- **Authentication & Security:** Implemented JWT token-based authentication, role-based access control (RBAC), and secure API endpoints with proper CORS configuration
- **Performance Optimization:** Implemented database query optimization, response caching, and efficient pagination for large datasets

Database Management (PostgreSQL):

- **Schema Design:** Designed normalized database schemas with proper foreign key relationships, indexes, and constraints for data integrity

- **Query Optimization:** Wrote efficient SQL queries with proper joins, subqueries, and aggregations for complex financial data analysis
- **Data Migration:** Implemented database migration scripts and version control for schema changes and data updates

Authentication & User Management (Firebase):

- **Firebase Authentication:** Integrated Firebase Auth for secure user registration, login, password reset, and social authentication options
- **Firestore Integration:** Utilized Firestore for real-time user data synchronization, company profile management, and user role assignments
- **Invite System:** Developed a secure invite-based user onboarding system where administrators can invite employees with role-specific permissions
- **User Roles:** Implemented role-based access control with different permission levels for administrators and employees

Map Integration & Geospatial Features:

- **Leaflet Maps:** Integrated OpenLeaf Maps (Leaflet) for interactive mapping functionality with custom markers, popups, and layer management
- **Google Places API:** Implemented Google Address autocomplete for enhanced user experience in property search and location-based queries
- **ATTOM API Integration:** Connected with ATTOM Data API to fetch comprehensive property information including market values, property details, neighborhood analytics, and investment metrics
- **Geospatial Analysis:** Implemented location-based filtering, radius searches, and geographic data visualization for investment opportunity mapping

Key Features Implemented:

- **Investment Analytics:** Real-time property valuation, market trend analysis, and investment opportunity scoring
- **Portfolio Management:** User bookmarking system, investment tracking, and performance analytics
- **Reporting System:** Automated report generation with PDF export capabilities for investment analysis
- **Data Visualization:** Interactive charts and graphs for market trends, property values, and investment performance metrics

Technical Achievements:

- Successfully handled concurrent user sessions and real-time data updates

- Implemented efficient caching strategies reducing API response times by 60%
 - Created scalable architecture supporting growth from MVP to production-ready application
 - Achieved 99.9% uptime during development and testing phases
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AI Plant Disease Detection System

Duration: March 2025 – April 2025

 **Technologies:** HTML5, CSS3, JavaScript ES6+, TensorFlow.js, Hugging Face Spaces, Python, CNN

Project Overview: Developed an intelligent plant disease detection system using deep learning to help farmers and agriculturists identify plant diseases from leaf images. The system provides accurate, real-time disease classification with detailed information about treatment recommendations and prevention strategies.

Machine Learning & AI Development:

- **Convolutional Neural Network (CNN):** Designed and trained a custom CNN architecture optimized for plant disease classification, implementing multiple convolutional layers, pooling layers, and dropout techniques for improved accuracy and generalization
- **Dataset Management:** Curated and preprocessed a comprehensive dataset of plant leaf images with various disease categories, implementing data augmentation techniques including rotation, scaling, brightness adjustment, and noise injection to improve model robustness
- **Model Training:** Implemented transfer learning using pre-trained models (ResNet, VGG, MobileNet) and fine-tuned them for plant disease classification, achieving high accuracy rates through systematic hyperparameter optimization
- **TensorFlow Implementation:** Built the model using TensorFlow 2.x with Keras API, implementing custom training loops, loss functions, and evaluation metrics specifically designed for multi-class plant disease classification

Model Optimization & Performance:

- **Architecture Optimization:** Experimented with different CNN architectures to balance accuracy and inference speed, implementing efficient convolution operations and optimized layer configurations
- **Quantization & Compression:** Applied model quantization techniques to reduce model size for web deployment while maintaining accuracy, making the model suitable for edge devices and mobile applications
- **Performance Metrics:** Achieved high classification accuracy with comprehensive evaluation using confusion matrices, precision, recall, F1-score, and ROC curves across different disease categories
- **Validation Strategy:** Implemented k-fold cross-validation and stratified sampling to ensure robust model evaluation and prevent overfitting

Web Application Development:

- **Frontend Development:** Created an intuitive, responsive web interface using modern HTML5 semantic elements, CSS3 with Grid and Flexbox layouts, and vanilla JavaScript ES6+ for interactive functionality
- **Image Processing:** Implemented client-side image preprocessing using JavaScript Canvas API, including image resizing, normalization, and format conversion for optimal model input
- **Real-time Inference:** Integrated TensorFlow.js for in-browser model inference, enabling real-time disease prediction without server-side processing
- **User Experience:** Designed an intuitive drag-and-drop interface with image preview, prediction confidence scores, and detailed disease information display

Deployment & Accessibility:

- **Hugging Face Spaces:** Deployed the complete application on Hugging Face Spaces, providing easy access and sharing capabilities for the agricultural community
- **Mobile Optimization:** Implemented mobile-first responsive design ensuring optimal user experience on smartphones and tablets, crucial for field usage by farmers
- **Cross-Platform Compatibility:** Ensured compatibility across different browsers and devices, implementing progressive web app features for offline capabilities
- **Performance Optimization:** Optimized loading times, implemented lazy loading for images, and minimized JavaScript bundle size for faster mobile performance

Agricultural Impact & Features:

- **Disease Classification:** Accurate identification of common plant diseases including bacterial blight, leaf spot, powdery mildew, and various fungal infections
- **Treatment Recommendations:** Provided detailed treatment suggestions, recommended fungicides, and preventive measures for each identified disease
- **Early Detection:** Enabled early-stage disease detection helping farmers take timely action to prevent crop loss
- **Educational Content:** Included educational information about disease symptoms, prevention strategies, and best agricultural practices

Technical Achievements:

- Achieved 94% accuracy in plant disease classification across 15+ disease categories
 - Processed and analyzed over 10,000 plant leaf images for training and validation
 - Implemented efficient model inference with sub-second prediction times
 - Created a scalable system that can be easily extended to include additional plant species and diseases
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Technical Skills & Expertise

Programming Languages

Python: Advanced proficiency in Python programming with extensive experience in web development (Django, Flask, FastAPI), data science (NumPy, Pandas, Matplotlib), machine learning (TensorFlow, PyTorch, scikit-learn), automation scripting, and API development. Skilled in object-oriented programming, functional programming paradigms, and Python best practices including PEP 8 compliance and unit testing.

JavaScript: Expert-level JavaScript development including ES6+ features, asynchronous programming (async/await, Promises), DOM manipulation, event handling, and modern JavaScript frameworks. Experience with both frontend (React, Vue.js) and backend (Node.js, Express.js) JavaScript development, along with TypeScript for type-safe development.

SQL: Proficient in writing complex SQL queries, database design, normalization, indexing strategies, stored procedures, and performance optimization. Experience with multiple database systems including PostgreSQL, MySQL, and MongoDB with expertise in query optimization, transaction management, and database security.

Java: Solid foundation in Java programming including object-oriented design principles, collections framework, exception handling, multithreading, and Java 8+ features. Experience with Spring framework, Maven/Gradle build tools, and Java-based web application development.

Web Technologies & Frameworks

Frontend Technologies:

- **HTML5:** Semantic markup, accessibility standards (WCAG), SEO optimization, and progressive web app development
- **CSS3:** Advanced styling including Grid, Flexbox, animations, transitions, responsive design, and CSS preprocessors (Sass/SCSS)
- **Bootstrap:** Responsive design framework, custom component development, and theme customization
- **Tailwind CSS:** Utility-first CSS framework, custom configuration, and component-based design systems

Frontend Frameworks:

- **React.js:** Component-based development, hooks, state management (Redux, Context API), lifecycle methods, and performance optimization
- **Vue.js:** Vue 3 with Composition API, Vuex state management, Vue Router, and component library development

Backend Technologies:

- **Node.js:** Server-side JavaScript, npm ecosystem, event-driven architecture, and asynchronous programming
- **Express.js:** RESTful API development, middleware implementation, authentication, and security best practices
- **FastAPI:** Modern Python web framework, automatic API documentation, dependency injection, and high-performance API development

Database Technologies

Relational Databases:

- **PostgreSQL:** Advanced SQL queries, indexing, partitioning, stored procedures, and database optimization
- **MySQL:** Database design, replication, backup strategies, and performance tuning

NoSQL Databases:

- **MongoDB:** Document-based database design, aggregation pipelines, indexing, and schema design for flexible data models
- **Neo4j:** Graph database modeling, Cypher query language, and relationship-based data analysis

Cloud & DevOps Tools

Amazon Web Services (AWS):

- **EC2:** Virtual server management, load balancing, auto-scaling, and instance optimization
- **S3:** Object storage, static website hosting, data backup, and content delivery
- **Lambda:** Serverless computing, event-driven architecture, and microservices development
- **IAM:** Identity and access management, security policies, and role-based access control

Development Tools:

- **Git:** Version control, branching strategies, merge conflict resolution, and collaborative development workflows
- **GitHub/GitLab:** Repository management, CI/CD pipelines, issue tracking, and project management
- **Docker:** Containerization, multi-stage builds, Docker Compose, and container orchestration
- **Figma:** UI/UX design, prototyping, design systems, and collaborative design workflows

Computer Science Fundamentals

Data Structures and Algorithms:

- **Data Structures:** Arrays, linked lists, stacks, queues, trees, graphs, hash tables, and heaps
- **Algorithms:** Sorting algorithms, searching algorithms, dynamic programming, greedy algorithms, and graph algorithms
- **Algorithm Analysis:** Time and space complexity analysis, Big O notation, and algorithm optimization
- **Problem Solving:** Competitive programming experience with 1000+ problems solved across various platforms

Operating Systems:

- Process management, memory management, file systems, and concurrent programming
- Linux system administration, shell scripting, and system optimization
- Understanding of system calls, multithreading, and inter-process communication

Database Management Systems:

- Database design principles, normalization, ACID properties, and transaction management
- Query optimization, indexing strategies, and database performance tuning
- Backup and recovery strategies, replication, and database security

Computer Networks:

- Network protocols (TCP/IP, HTTP/HTTPS, DNS), network architecture, and security
- RESTful API design, web services, and distributed systems
- Network troubleshooting and performance optimization

Object-Oriented Programming:

- OOP principles (encapsulation, inheritance, polymorphism, abstraction)
- Design patterns (Singleton, Factory, Observer, MVC) and software architecture
- Code quality, testing strategies, and maintainable code practices

Professional Certifications

Build Your Own Dynamic Web Application

Issuing Organization: NxtWave Academy



Completion Date: [Date]



Skills Acquired: Advanced JavaScript ES6+ concepts, DOM manipulation, asynchronous programming, fetch API, local storage, and modern web development practices. Learned to build responsive, interactive web applications with dynamic content loading, user authentication, and state management.

Programming Foundations with Python

Issuing Organization: NxtWave Academy



Completion Date: [Date]



Skills Acquired: Comprehensive Python programming including object-oriented programming principles, data structures (lists, dictionaries, sets), file handling, exception handling, and real-world application development. Gained expertise in writing clean, efficient Python code and understanding programming best practices.

Introduction to SQL

Issuing Organization: NxtWave Academy



Completion Date: [Date]



Skills Acquired: Mastered SQL fundamentals including database querying, data filtering, joins (inner, outer, cross), subqueries, aggregate functions, and data manipulation. Learned advanced SQL concepts like window functions, stored procedures, and database optimization techniques.

AWS Cloud Fundamentals

Issuing Organization: NxtWave Academy



Completion Date: [Date]



Skills Acquired: Comprehensive understanding of Amazon Web Services including EC2 instance management, S3 object storage, Lambda serverless functions, IAM security policies, and basic cloud architecture patterns. Gained hands-on experience with cloud deployment, monitoring, and cost optimization strategies.

Achievements & Recognition

Competitive Programming Excellence

- **LeetCode Global Ranking:** Achieved Global Rank 1599 in LeetCode Weekly Challenge among 19,000+ participants worldwide, demonstrating exceptional problem-solving skills and algorithmic thinking
- **Problem Solving Expertise:** Successfully solved 1000+ coding problems across multiple platforms including LeetCode, HackerRank, CodeChef, and Codeforces, showcasing consistency and dedication to continuous learning
- **Algorithm Mastery:** Demonstrated proficiency in various algorithmic concepts including dynamic programming, graph algorithms, tree traversals, and complex data structure implementations

Hackathon & Competition Success

- **Fulued 2.0 Contest:** Secured Top 50 position among 5000+ participants in a highly competitive technical challenge, demonstrating practical problem-solving skills and ability to perform under pressure

- **AI HACK DAYS 2025:** Finalist in a prestigious AI hackathon, showcasing innovation in artificial intelligence, machine learning implementation, and rapid prototype development
- **Brandquezt 21:** Achieved Global Rank 200 in "Brands Behind the Scenes" competition among 25,000+ participants, demonstrating business acumen and analytical thinking

Industry Recognition

- **Naukri Campus Young Turks:** Top 1% Performer in India's Largest Skill Contest among 500,000+ participants, recognized for exceptional technical skills, programming ability, and overall technical competency
- **National Level Recognition:** Consistent performance in national-level coding competitions and technical challenges, establishing reputation as a skilled programmer and problem solver

Technical Leadership & Community Impact

- **Open Source Contributions:** Active contributor to open-source projects with focus on web development, AI/ML tools, and educational resources
 - **Mentorship:** Provided guidance and mentorship to junior developers and students in programming, web development, and career development
 - **Knowledge Sharing:** Regular participant in technical discussions, code reviews, and knowledge-sharing sessions within development communities
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Additional Information

Professional Interests

- **Artificial Intelligence & Machine Learning:** Passionate about developing AI solutions for real-world problems, particularly in natural language processing, computer vision, and regional language technologies
- **Full-Stack Development:** Enthusiastic about building end-to-end web applications that provide seamless user experiences and solve business challenges
- **Cloud Computing:** Interested in scalable cloud architectures, serverless computing, and modern DevOps practices

Personal Qualities

- **Problem-Solving Mindset:** Approach challenges with analytical thinking and systematic problem-solving methodologies
- **Continuous Learning:** Committed to staying updated with latest technologies, industry trends, and best practices through continuous learning and professional development
- **Team Collaboration:** Strong communication skills and ability to work effectively in cross-functional teams and diverse cultural environments

- **Quality Focus:** Dedication to writing clean, maintainable code and following software development best practices

Languages

- **English:** Fluent (Professional working proficiency)
- **Telugu:** Native speaker
- **Hindi:** Conversational proficiency

Availability

- **Current Status:** Available for full-time opportunities starting [Date]
- **Preferred Roles:** Software Developer, Full-Stack Developer, AI/ML Engineer, Backend Developer
- **Location Preference:** Open to opportunities in Hyderabad, Bangalore, or remote work arrangements